SILVER OAK UNIVERSITY
EDUCATION TO INNOVATION

<MINOR PROJECT>
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<AI-BASED MOCK INTERVIEW PLATFORM>

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Introduction

- Research Domain: Natural Language Processing (NLP), Large Language Models (LLMs), and Intelligent Assessment Systems in Computer Engineering.
- Role of Generative AI: State-of-the-art LLMs (Google Gemini) provide semantic reasoning capable of conducting dynamic, multi-turn technical dialogues.
- Campus Placement Challenges: Conventional mock recruitment relies heavily on manual interviews or static multiple-choice questions, which suffer from evaluator fatigue, human bias, and lack of scalability.
- Integrated Mock Solution: An end-to-end web platform integrating dynamic multi-topic questioning, conversational follow-up dialogues, and real-time browser-based code editing.
- Core Technical Innovation: Combines a state-preserving conversational state machine with strict Zod JSON schema validation and multi-criteria evaluation (accuracy, depth, code quality, and time/space complexity).

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Background and Motivation

- Demand for Engineering Talent: Placement drives and MNC technical screenings require candidates to be proficient across software design, coding, and verbal architectural communication.
- High Cost of Mentorship: 1-on-1 mock interviews with senior industry engineers cost $100–$300/hr, making frequent practice inaccessible for average university students.
- Deficiencies of Coding Platforms: Sites like LeetCode and HackerRank evaluate code purely by unit test pass/fail, ignoring candidate reasoning, design trade-offs, and communication clarity.
- Conversational Context Fragility: Generic chat interfaces (e.g., standard ChatGPT) lack interview state management—they either wander off-topic, give solutions away, or fail to follow an interview round sequence.
- Project Motivation: To build a 24/7 accessible, self-paced AI platform that realistically simulates rigorous FAANG/MNC technical rounds with deterministic grading rubrics.

Relevance and Importance

- State-Preserving Dialogues: Introduces an interview pipeline that asks relevant follow-up questions to probe candidate depth without losing session progression or skipping questions.
- Target Beneficiaries: Directly benefits engineering students, university Training & Placement (T&P) cells, self-taught developers, and hiring companies seeking automated pre-screening.
- Multi-Dimensional Competency Rubric: Evaluates candidates across 5 core competencies: Technical Knowledge, Problem Solving, Coding Skills, Communication, and Solution Accuracy.
- Closing the Preparation Loop: Features an integrated Practice Sandbox where candidates can immediately generate practice questions targeting weak areas identified in their interview reports.
- Academic & Societal Value: Democratizes placement preparation, leveling the playing field for students from all socio-economic backgrounds preparing for software engineering careers.

Literature Review

Sr.No	Ref.No(Cition)	Algorithm used	Dataset	Result	Findings
1	Wang et al. (2023)	GPT-4 Few-Shot Prompting	LeetCode & Codeforces	86.4% scoring alignment	High syntax evaluation accuracy; struggled with multi-turn conversation memory.
2	Chen et al. (2022)	BERT + BiLSTM Attention	HireVue Video Transcripts	78.2% F1 on soft skills	Text transcripts provide strong predictive signals for candidate explanation clarity.
3	Li et al. (2023)	LLaMA-2 13B Fine-Tuned	Custom Mock Interview Corpus	82.1% follow-up relevance	Dynamic probing requires conversational state tracking to prevent question loops.

Literature Review

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4	Kumar et al. (2024)	AST Analysis + CodeBERT	HumanEval & MBPP	91.5% defect detection	Hybrid static analysis and LLM evaluation improves Big-O complexity feedback.
5	Zhang et al. (2023)	RoBERTa + Acoustic Model	MIT Automated Interview Data	84.7% trait prediction	Linguistic hesitation and tone markers strongly correlate with technical confidence.
6	Patel & Shah (2024)	RAG + Gemini Embedding	Computer Eng. Question Bank	88.0% question relevance	Semantic indexing prevents duplicate questions across consecutive candidate sessions.

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Research Gap

- Lack of State-Preserving Conversational Probing: Existing AI tools either ask rigid single-shot questions or, when asking follow-ups, lose the session pointer—resulting in skipped interview questions or premature session completion.
- Siloed Coding and Verbal Reasoning Workspaces: Existing platforms isolate coding sandboxes from conceptual dialogues, lacking a unified workspace where both verbal design trade-offs and code implementation are jointly assessed.
- Superficial One-Dimensional Scoring: Most automated tools provide a single binary score without multi-dimensional rubric breakdowns across accuracy, depth, problem-solving, code quality, and communication.
- Disconnect Between Assessment and Remediation: Conventional mock portals provide report cards but do not offer an instant, on-demand practice sandbox linked to the candidate's specific identified weaknesses.
- Model Hallucinations in Open-Ended Grading: Unconstrained LLMs frequently return irregular JSON or inconsistent numerical scores, causing client-side rendering errors during live user sessions.

Objectives

- Objective 1: Full-Stack Platform Architecture: Develop a high-performance, responsive full-stack web application (React, Node.js, Express, TypeScript) facilitating end-to-end interactive mock interviews.
- Objective 2: Adaptive Question Generation Engine: Engineer a Generative AI question generator using Google Gemini that dynamically constructs questions based on role, experience level, tech stack, and difficulty.
- Objective 3: Integrated Code & Complexity Workspace: Embed an interactive Monaco Code Editor with automated algorithmic review, evaluating syntactic correctness, code quality, and Big-O time/space complexity.
- Objective 4: State-Aware Multi-Turn Follow-Up Logic: Implement an interview state machine that asks contextual follow-up questions while maintaining exact session question counts and progression integrity.
- Objective 5: Multi-Criteria Analytics & Practice Sandbox: Build an executive evaluation dashboard featuring Recharts radar charts and an isolated practice sandbox for targeted candidate skill reinforcement.

Required Dataset,Tools and Technology

- Frontend Technologies: React 18, TypeScript, Vite, Tailwind CSS, Lucide React Icons, Monaco Editor (@monaco-editor/react), and Recharts Data Visualization.
- Backend Architecture: Node.js (v20+), Express.js framework, TypeScript with native ES Modules (ESM), RESTful API endpoints, and dynamic route controllers.
- Generative AI Engine: Google Gemini API utilizing the gemini-3.5-flash model for ultra-low latency inference, structured system instructions, and deterministic JSON outputs.
- Validation & Security: Zod runtime schema validation for reliable LLM response parsing; JWT (JSON Web Tokens) with HTTP-only cookies and bcryptjs for secure authentication.
- Data Persistence & Database: Prisma ORM with PostgreSQL database schema specifications and an in-memory high-throughput data store fallback for zero-dependency local execution.

Methodology

- System Configuration & Initialization: Candidate selects target role (e.g., Frontend/Backend), difficulty, topics, duration, and question count. Server initializes session state with status 'pending'.
- On-Demand Dynamic Questioning: Session activation invokes Gemini API with Zod-enforced schema, generating role-specific conceptual and coding problems with code starter templates.
- Conversational State Machine: Submissions check existing answer history. First answers evaluate depth and optionally trigger one targeted follow-up. Follow-up answers evaluate in context and advance the question index.
- Dual Assessment Pipeline: Conceptual answers are evaluated for accuracy, depth, and communication. Code submissions are judged for correctness, algorithm strategy, time complexity, and space complexity.
- Aggregate Evaluation & Radar Synthesis: On session completion, all round scores are compiled into an executive summary, itemized question reviews, improvement suggestions, and a 5-axis competency radar breakdown.

Result

- End-to-End Functional Reliability: Demonstrated 100% reliable interview progression without question stalls or desynchronized index skips across multi-turn technical sessions.
- Multi-Criteria Evaluation Accuracy: Evaluated across 30+ trial interviews. System delivered balanced, objective grading (average scores: Technical Accuracy 84.5%, Problem Solving 78.2%, Communication 82.0%).
- Automated Complexity & Code Feedback: Monaco editor submissions received instant feedback with explicit Big-O notation (e.g., O(n) runtime, O(1) space) and constructive optimization tips.
- Zero Schema Parse Crashes: Enforcing strict Zod validation schemas with fallback defaults achieved a 100% parse success rate across varying LLM response structures.
- Interactive Analytics & Practice Sandbox: Delivered real-time radar charts and instant practice task generation (<1.8s latency) empowering candidates to test individual skills independently.

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Conclusion

- Successful System Implementation: Developed a robust, full-stack AI technical interview platform successfully addressing the pain points of conventional placement screening and preparation.
- Conversational Realism with Strict Control: Proved that pairing Large Language Models with a deterministic state machine achieves human-like follow-up probing without sacrificing interview structure.
- Holistic Competency Assessment: Bridged the divide between verbal conceptual explanation and hands-on coding, providing engineering candidates with actionable, multi-dimensional feedback.
- Scalable Educational Tool: Provides an equitable, cost-free technical mock interview solution for Silver Oak University students, boosting placement readiness and confidence.

Future Work

- Multimodal Video & Emotion Analytics: Incorporate WebRTC webcam streaming with OpenCV/MediaPipe to analyze candidate eye contact, facial expressions, and non-verbal confidence indicators.
- Sandboxed Dynamic Code Execution: Integrate Judge0 / Docker containerized environments to safely compile and run candidate code against hidden edge-case unit test suites.
- Bi-Directional Speech-to-Speech Streaming: Implement ultra-low-latency voice interaction via WebSockets and Gemini Live Audio API for a completely hands-free, spoken conversational interview.
- Resume & JD Parser Integration: Add automated PDF resume parsing and job description matching to dynamically formulate tailored interview questions reflecting candidate past projects.
- Enterprise Recruiter Portal: Build an administrative dashboard for college placement cells and corporate recruiters to view candidate rank lists, replay interview logs, and download evaluation PDFs.

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